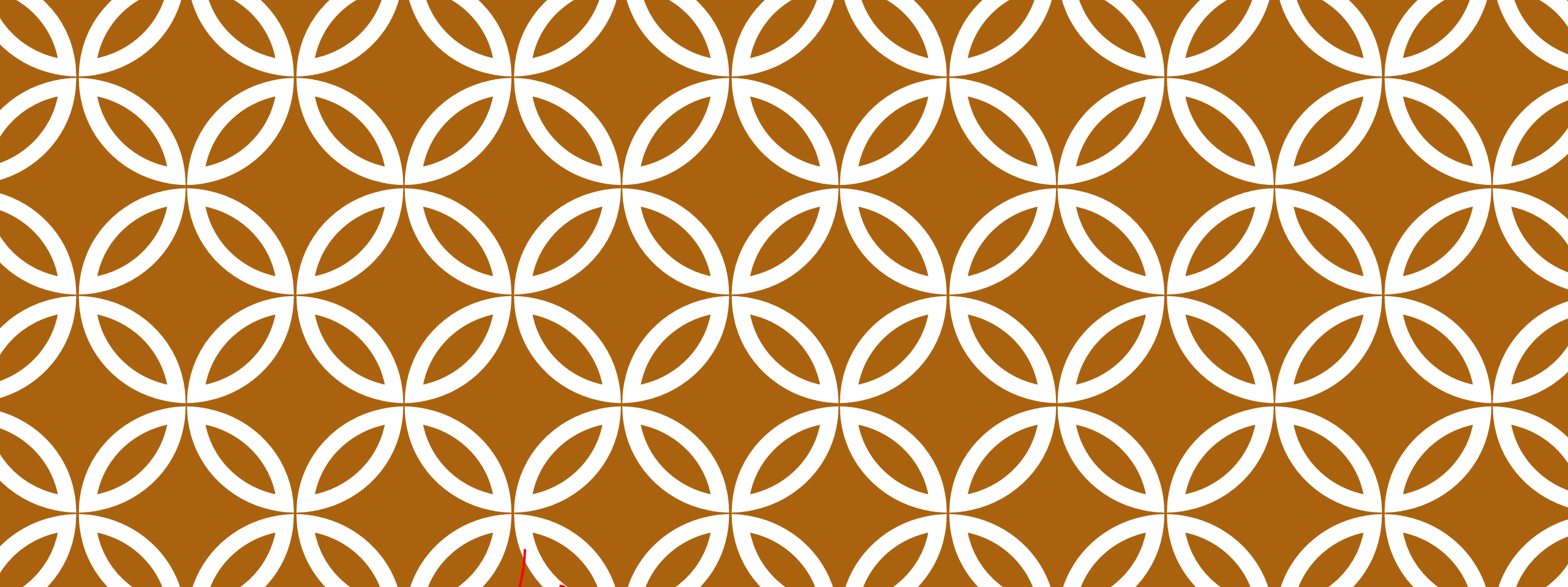




Simplify / Reducing

MINIMIZE BOOLEAN EXPRESSIONS - 2

Unit – 2: Digital Electronics

Limitations using Boolean Laws

- ▶ If we minimize Boolean functions using Boolean postulates and theorems.
- ▶ It is a time consuming process and re-write the simplified expressions after each step.
- ▶ Although, that method was helpful but also was difficult and non-systematic. *lengthy*
- ▶ We don't have any guarantee that we have reached a minimal expression.
- ▶ Thus, Karnaugh Maps (K-Maps) are used to simplify Boolean functions. *2 → 5 variable*
- ▶ K-Map is a systematic graphical method to simplify Boolean expressions. *2-D*

K-MAP

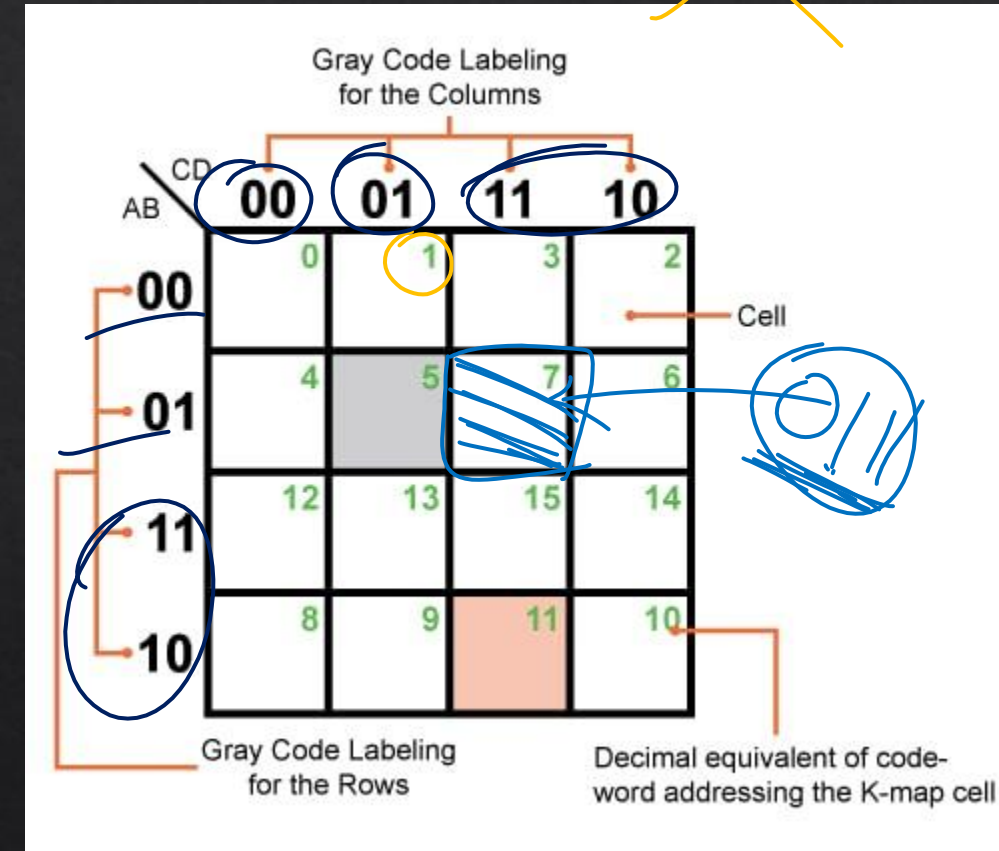
- A Karnaugh map (K-Map) is defined as a visual tool used to simplify Boolean expressions by organizing values into a grid. *Table*
- A K-Map is a 2D chart or graph composed of an arrangement of adjacent cells.
- K-maps basically deals with the technique of inserting the values of the output variable in cells within a rectangle or square grid according to a definite pattern.
- In a truth table, a function of n variables will have 2^n min-terms. *2 to the power of n*
- For example, there will be four min-terms for two variables, i.e. (x,y), and 8 min-terms for 3 variables. *2^2 = 4*
- The adjacent cells in K-Map are differed only in single bit position. *2^3 = 8*

~~0~~
~~1~~
~~1~~
~~1~~

~~current~~
~~next~~

~~0~~
~~1~~
~~1~~
~~1~~

- [illegible]



K-MAP SIMPLIFICATION PROCESS

1. Create the K-Map based on the number of variables (2, 3, 4, or more).
2. Plot the given Boolean function into the K-Map (place 1's for SOP or 0's for POS).
3. Group adjacent 1's (for SOP) or 0's (for POS) in powers of 2 (1, 2, 4, 8...).
 - ✓ Group must be rectangular or square.
 - ✓ Groups can wrap around the edges.
4. Write the simplified expression based on groups.
5. Combine the simplified terms to get the final minimized expression.


$$F(x, y) = \sum m(0, 3) = \overline{x'y'} + \overline{xy}$$

IMPORTANT GROUPING RULES

- Groups must contain 1, 2, 4, 8, 16... cells (powers of 2).

0	1	1	1
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Incorrect

0	1	1	1
---	---	---	---

Correct

- Each group should be as large as possible.

1	1	1	1
1	1	1	1

Incorrect

1	1	1	1
1	1	1	1

Correct

- Each 1 (or 0) must be in at least one group.
- Overlap is allowed (a 1 or 0 can be part of multiple groups).

1	1	1	1
0	1	1	0

Incorrect

1	1	1	1
0	1	1	0

Correct

- Grouping is to be done either horizontally or vertically or in terms of squares or rectangles. Diagonal grouping of 'ones' is not permitted.

0	1	0	0
0	1	1	0

Incorrect

0	1	0	0
0	1	1	0

Correct

- The elements around the edges of the table are considered to be adjacent and can be grouped together.

1	0	0	1
1	0	0	1

- Don't care conditions*** are to be considered only if they aid in increasing the group-size (else neglected).

1	1	1	1
X	1	X	0

Neglect Consider

2-VARIABLE K-MAP

$2^2 = 4 \leftarrow \text{cell} \leftarrow \text{table}$

- ✓ A 2-variable expression can have $2^2 = 4$ possible ways of input representation.
- ✓ Thus, a 2-variable K-Map will have 4 adjacent cells representing each of the 4 possible combinations of inputs.
- ✓ Binary numbers at the top of the K-Map represent the condition of variable B along any column and,
- ✓ Binary number to the left of the K-Map represents the condition of variable A along any row.



◆ Representation using minterms (SOP form)

$$0 \Rightarrow A' \text{ or } B'$$

$$1 \Rightarrow A \text{ or } B$$

A \ B	0	1
0	m_0 $\bar{A} \cdot \bar{B}$	m_1 $\bar{A} \cdot B$
1	m_2 $A \cdot \bar{B}$	m_3 $A \cdot B$

minterm
index

- ◆ The possible combinations of grouping 2 adjacent min terms are $\{(m_0, m_1), (m_2, m_3), (m_0, m_2)$ and $(m_1, m_3)\}$.

◆ Representation using maxterms (POS form)

$$0 \Rightarrow A \text{ or } B$$

$$1 \Rightarrow A' \text{ or } B'$$

A \ B	0	1
0	M_0 $\bar{A} + \bar{B}$	M_1 $\bar{A} + B$
1	M_2 $A + \bar{B}$	M_3 $A + B$

A	B	max index
0	0	M_0
0	1	M_1
1	0	M_2
1	1	M_3

- ◆ The possible combinations of grouping 2 adjacent min terms are $\{(M_0, M_1), (M_2, M_3), (M_0, M_2)$ AND $(M_1, M_3)\}$.

EXAMPLE: REDUCE BOOLEAN FUNCTION $F(A, B) = \sum m(1, 3)$ USING K-MAP

Meaning F is 1 for minterms 1 and 3.

✓ There are two 1's vertically aligned in column B=1.

✓ Group them together.

✓ In this column, B=1 is constant, but A changes (0 → 1).

So, eliminate it.

✓ Therefore, the simplified term is: $F = \underline{B}$

So, 4 cell

$\backslash B$	0	1
A 0	0	1
A 1	0	1

Group of 2

Result: $F = B$

Group

$$m_1 = A'B$$
$$m_3 = AB$$

$$f(A, B) = m_1 + m_3 = A'B + AB = B$$

EXAMPLE: REDUCE THE FUNCTION $F(A, B) = \sum m(0, 2, 3)$ USING K-MAP

Meaning F is 1 for minterms 0, 2 and 3

$\Rightarrow m_0, m_2 \& m_3$

$\Rightarrow SOP$

Group-1 (Vertical Pair)

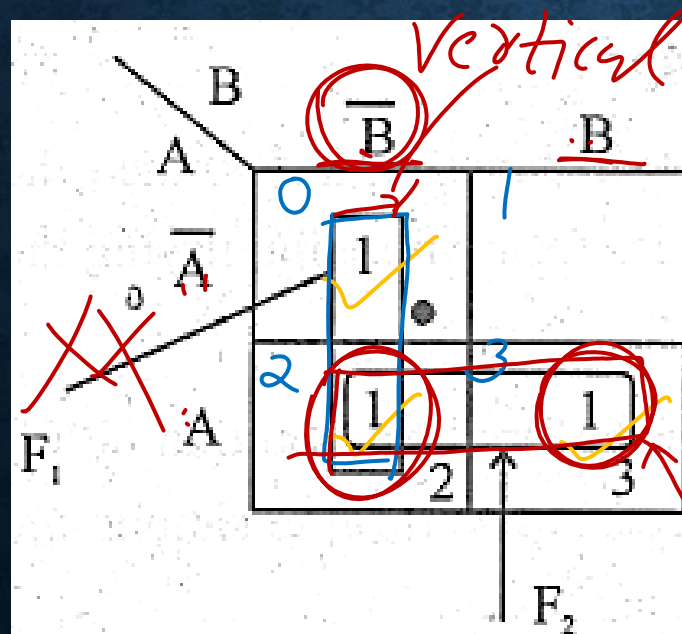
✓ Cells: $m_0(00)$ and $m_2(10)$

✓ $B = 0$ remains constant $= B'$

Group-2 (Horizontal Pair)

✓ Cells: $m_2(10)$ and $m_3(11)$

✓ $A = 1$ remains constant $= A$



Put the one in the given minterms

$$F = F_1 + F_2$$

$$F_1 = \bar{B}, F_2 = A$$

$$F = \bar{B} + A$$

$$\Rightarrow F(A, B) = A + B'$$



Example: Simplify the Boolean Function:

$F(x, y) = xy' + x'y + xy$ using K-Map

Here, minterms xy' : (1, 0) = m_2 , $x'y$: (0, 1) = m_1 and xy : (1, 1) = m_3

So, $F(x, y) = \sum m(1, 2, 3)$

Group-1 (Horizontal Pair)

Cells: $m_2(10)$ and $m_3(11)$.

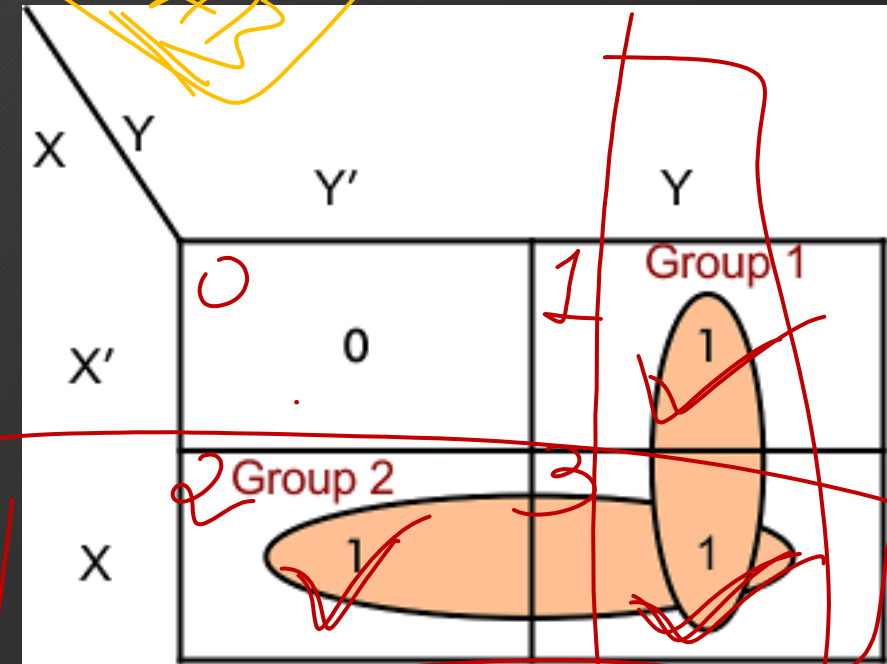
$x = 1$ constant = x

Group-2 (Vertical Pair)

Cells: $m_1(01)$ and $m_3(11)$

$y = 1$ constant = y

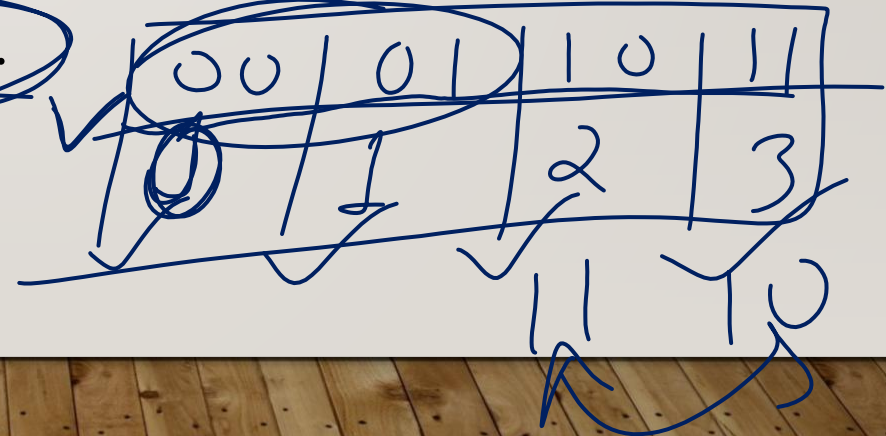
Simplified Expression: $F(x, y) = x + y$



$$F(x, y) = xy' + x'y + xy = A'B + A'B' + AB + AB' = A + B$$

3-VARIABLE K-MAP

- A 3-variable expression can have $2^3 = 8$ possible ways of input representation.
- Thus, a 3-variable K-Map will have 8 adjacent cells representing each of the 8 possible combinations of inputs.
- The thing to be noticed in the representation is that the binary numbers along the top of the K-Map are not in normal binary order.
- They are, in fact, in the Gray Code.



Representation using minterms (SOP form)

Handwritten notes: 2x4, 12, 3, 4

A \ BC	00	01	11	10
0	$\bar{A}\bar{B}\bar{C}$ 0	$\bar{A}\bar{B}C$ 1	$\bar{A}B\bar{C}$ 3	$\bar{A}BC$ 2
1	$A\bar{B}\bar{C}$ 4	$A\bar{B}C$ 5	$AB\bar{C}$ 7	ABC 6

A \ BC	00	01	11	10
0	m_0 0	m_1 1	m_3 3	m_2 2
1	m_4 4	m_5 5	m_7 7	m_6 6

100

~~101~~ ~~111~~ ~~110~~

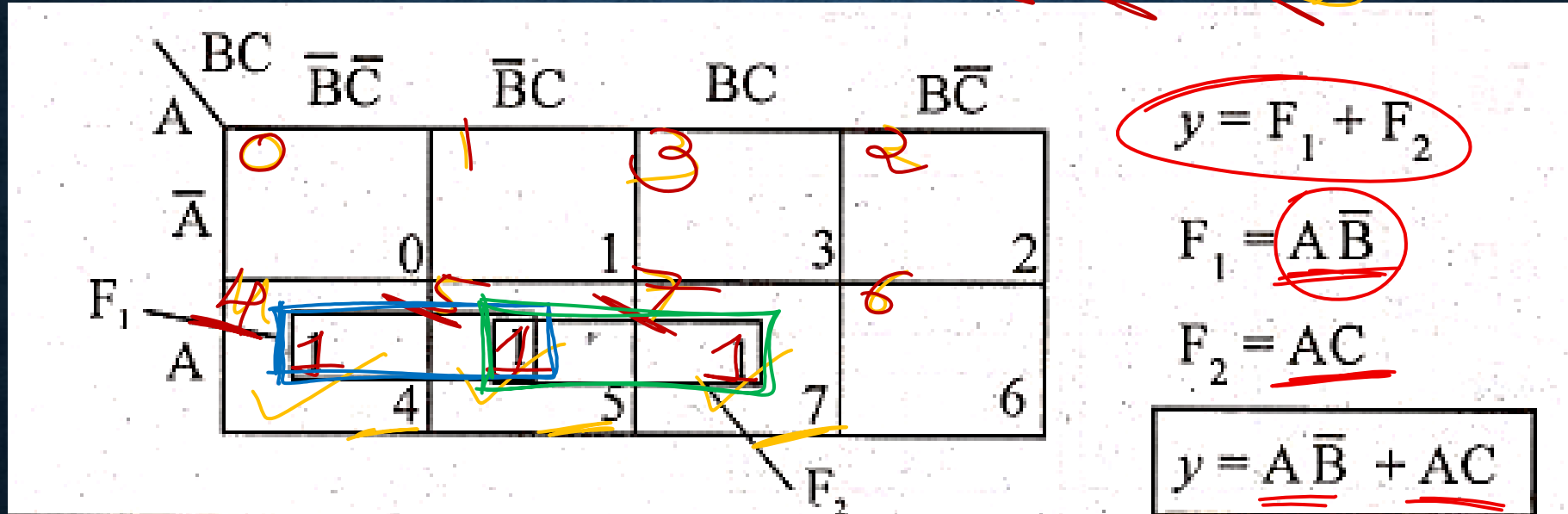
Representation using maxterms (POS form)

Handwritten notes: 0, 1, 3, 2

A \ BC	00	01	11	10
0	$A+B+C$	$A+B+\bar{C}$	$A+\bar{B}+\bar{C}$	$A+\bar{B}+C$
1	$\bar{A}+B+C$	$\bar{A}+B+\bar{C}$	$\bar{A}+\bar{B}+\bar{C}$	$\bar{A}+\bar{B}+C$

A \ BC	00	01	11	10
0	M_0 0	M_1 1	M_3 3	M_2 2
1	M_4 4	M_5 5	M_7 7	M_6 6

EXAMPLE: REDUCE THE FUNCTION $Y = \Sigma(4,5,7)$ USING K-MAP

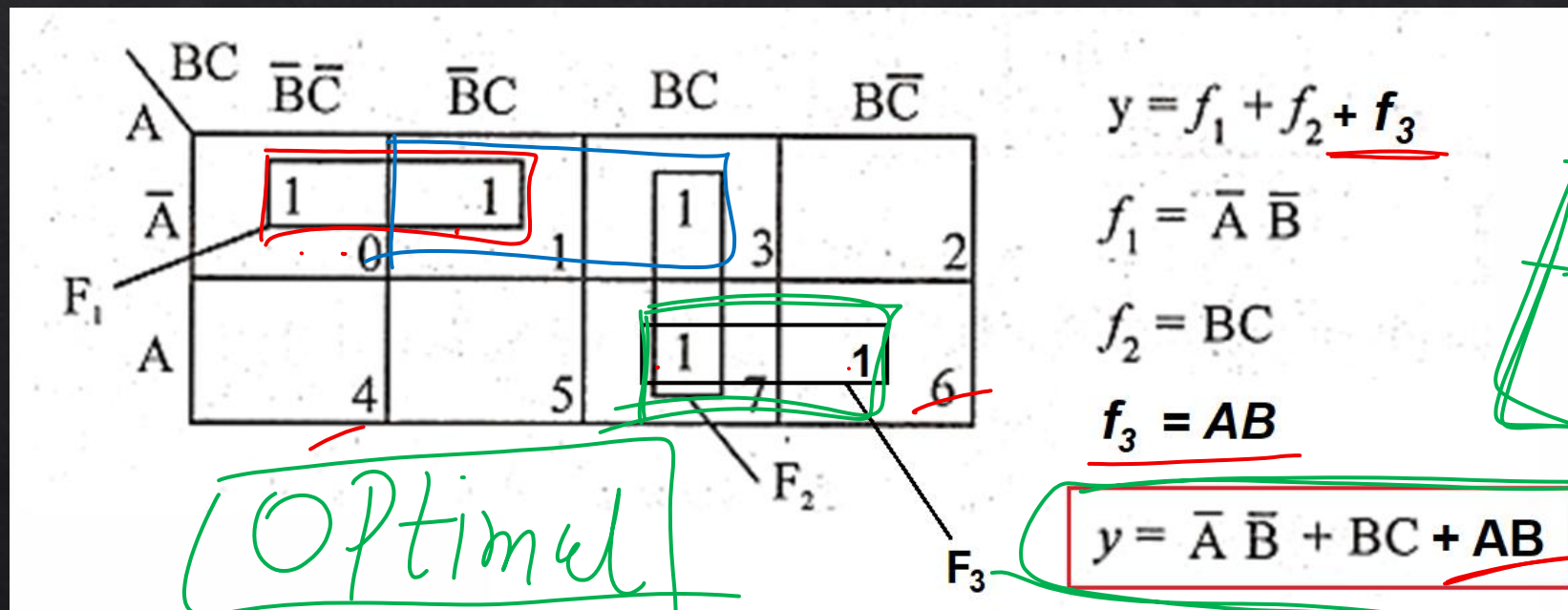


$$\{\underline{m_4, m_5}\} \& \{m_5, m_7\}$$

Example: Reduce the function $y(A,B,C) = \Sigma (0,1,3,6,7)$ using K-Map

- ◆ Here the 3 variables as in the function so we draw 3 variable k-map and fill ones in the given minterms: m_0, m_1, m_3, m_6 and m_7 .

Group 1: m_0, m_1 $A = 0, B = 0$ (C varies) $\Rightarrow A'B'$	Group 2: m_3, m_7 $B = 1, C = 1$ (A varies) $\Rightarrow BC$	Group 2: m_6, m_7 $A = 1, B = 1$ (C varies) $\Rightarrow AB$
--	--	--



~~AND = 3~~
~~OR = 2~~
~~NOT = 2~~

SIMPLIFY THE SOP FUNCTION $F(A,B,C) = \Sigma (1,2,3,7)$ USING K-MAP

	BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
A					
$\bar{A}$			1	1	1
A			1		

$F_1 = \bar{A}C$ (grouping m1, m3, m5, m7)
 $F_2 = \bar{A}B$ (grouping m1, m2, m5, m6)
 $F_3 = BC$ (grouping m3, m7)

$F = F_1 + F_2 + F_3$
 $F = \bar{A}C + \bar{A}B + BC$

Without groups

SIMPLIFY THE SOP FUNCTION $F(A,B,C) = \Sigma (1,3,5,7)$ USING K-MAP

Grouping:

- **Vertical check:** The grouping spans both rows where $A=0$ and $A=1$. Since A changes, it is eliminated.

- **Horizontal check:** The grouping spans columns where BC is 01 and 11.

✓ B changes from 0 to 1, so it is eliminated.

✓ C stays constant at 1.

- **Final Simplified Expression:** Because A and B both change across this quad, **only C remains.**

	BC	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$\overline{B}\overline{C}$
A			1	1	
A		0	1	3	2
A			1	1	
A		4	5	7	6

$$y = c$$

Write proper equation from given K-map.